
◆ AWS CLOUD PRACTITIONER BOOTCAMP

WEEK 1 — SESSION 1

CLOUD FOUNDATIONS & CORE SERVICES

Student Reference Notes (Premium Edition)

◆ 1. CLOUD COMPUTING — CORE IDEA

Cloud computing is the **delivery of IT resources over the internet on demand**, without owning physical infrastructure.

Key Characteristics:

- On-demand provisioning
- No upfront hardware investment
- Pay only for what you use
- Global accessibility

Conceptual Shift:

From **owning infrastructure** → to **consuming services**

◆ 2. CLOUD vs ON-PREMISE

Cloud (AWS)	On-Premise
No hardware ownership	Requires physical servers
Instantly scalable	Limited scaling capacity
Pay-as-you-go	High upfront cost
Managed infrastructure	Full responsibility

Key Insight:

Cloud removes operational burden and enables rapid scaling.

◆ 3. CAPEX vs OPEX (BUSINESS VIEW)

CapEx (Capital Expenditure)

- Large upfront investment
- Long-term asset ownership
- Fixed capacity

OpEx (Operational Expenditure)

- Pay per usage
- No ownership
- Flexible and scalable

AWS Model:

AWS operates primarily on **OpEx**, enabling cost efficiency and agility.

◆ 4. SHARED RESPONSIBILITY MODEL

Security in AWS is a **shared responsibility**.

AWS is responsible for:

- Physical infrastructure
- Data centers
- Hardware & networking

Customer is responsible for:

- Data protection
- Identity and access management
- Application configuration

Key Principle:

AWS secures the cloud

You secure what you put in the cloud

◆ 5. AWS GLOBAL INFRASTRUCTURE

AWS operates at a global scale.

Components:

Regions

- Physical geographic locations (e.g., Africa, Europe)

Availability Zones (AZs)

- Multiple isolated data centers within a region
- Designed for high availability

Edge Locations

- Used for fast content delivery globally

Key Principle:

Cloud architecture is designed for **global resilience**

◆ 6. REGIONS vs AVAILABILITY ZONES

- A **Region** contains multiple **Availability Zones**
- AZs are **isolated but connected**

Why this matters:

- Enables fault tolerance
- Prevents single points of failure

Exam Insight:

High availability = **Multi-AZ deployment**

◆ 7. CORE AWS SERVICE CATEGORIES

All AWS solutions are built on three pillars:

Compute

Processing power (running applications)

Storage

Saving and managing data

Security

Controlling access and protecting resources

◆ 8. CORE SERVICES OVERVIEW

◆ Amazon EC2 (Compute)

- Virtual servers in the cloud
- Full control of operating system
- Scalable on demand

Use Case:

Running applications, hosting servers

Exam Trigger:

“Need a server” → EC2

◆ Amazon S3 (Storage)

- Object-based storage system
- Stores files, images, backups
- Extremely durable

Use Case:

File storage, backups, static content

Exam Trigger:

“Store files/data” → S3

◆ AWS IAM (Security)

- Manages users and permissions
- Controls access to AWS services
- Uses roles and policies

Use Case:

Access control and security management

Exam Trigger:

“Who can access what?” → IAM

◆ 9. HOW CORE SERVICES WORK TOGETHER

Basic AWS architecture:

- EC2 → Runs applications
- S3 → Stores data
- IAM → Controls access

Concept:

These services form the **foundation of most AWS solutions**

◆ 10. REAL-WORLD APPLICATION

Example: Hosting a Website

- EC2 → Hosts the web server
- S3 → Stores images and assets
- IAM → Secures access and permissions

Insight:

This is a simplified version of real production systems.

◆ 11. QUICK RECALL (EXAM MEMORY MAP)

- EC2 → Compute
- S3 → Storage
- IAM → Security

Instruction:

Memorize these mappings—they appear frequently in exam scenarios.

◆ 12. SESSION OUTCOME

After this session, you should be able to:

- Explain the concept of cloud computing
 - Differentiate cloud vs on-premise
 - Understand AWS global infrastructure
 - Identify core AWS services and their use cases
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◆ FINAL PRINCIPLE

AWS is not about memorizing services.

It is about **understanding patterns and selecting the right tool for the scenario.**

◆ HOW TO USE THESE NOTES

- Review within 24 hours
 - Focus on **service** → **use case mapping**
 - Practice identifying services in scenarios
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